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TITLE PAGE

- **Problem Statement ID – 25236**
- **Problem Statement Title – Application software to obfuscate the object file using Low Level Virtual Machine (LLVM)**
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID – 89599**
- **Team Name (Registered on portal) – ByteRoots**

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HAICO

Hybrid AI-Enabled Code Obfuscator

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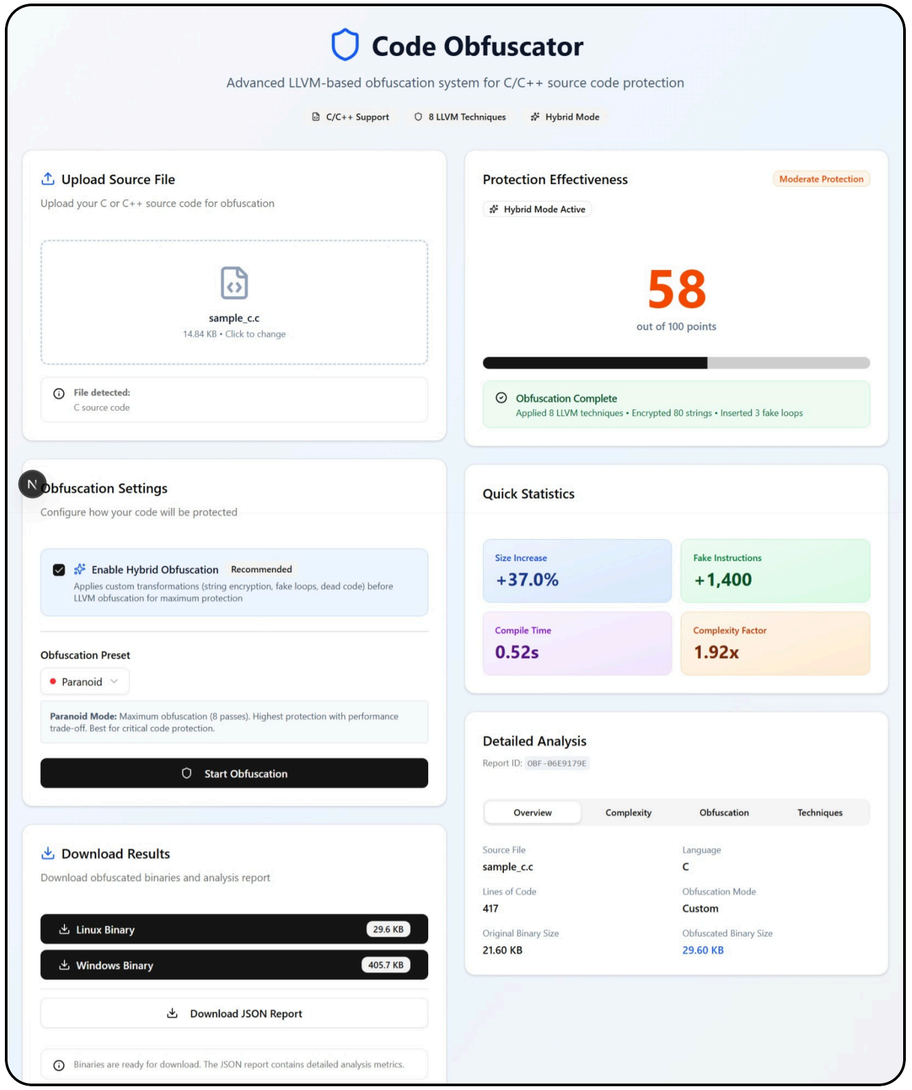
Proposed Solution

- **THREE Layer Obfuscation Architecture –**
  - Layer **1** – **Gen AI code snippet reconstruction** with **equivalent** but **confusing code**.
  - Layer **2** – **Custom Source Code Transformation** to encrypt strings, add fake loops, bogus code, and function changes.
  - Layer **3** – **LLVM IR & MIR Level Obfuscation** using **12+ techniques** (string encryption, indirect calls, dirty bytes, function splitting, etc) using Polaris.
- Analysis & Report –
  - **Comparing original & obfuscated binary objects.**
  - **Obfuscation effectiveness score** display based on parameters.
  - **Exact metrics tracking** and **Display with comparison**.
- Web Application Interface **(Full GUI Based) –**
  - **Upload C/C++ files** & **Download Binaries (Windows/Linux)** and **JSON analysis report**.
  - Backend VM with **sandbox environment** for **file processing** & **compilation**.

Demo Video – [Click Here]



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WORKFLOW AND TECH STACK

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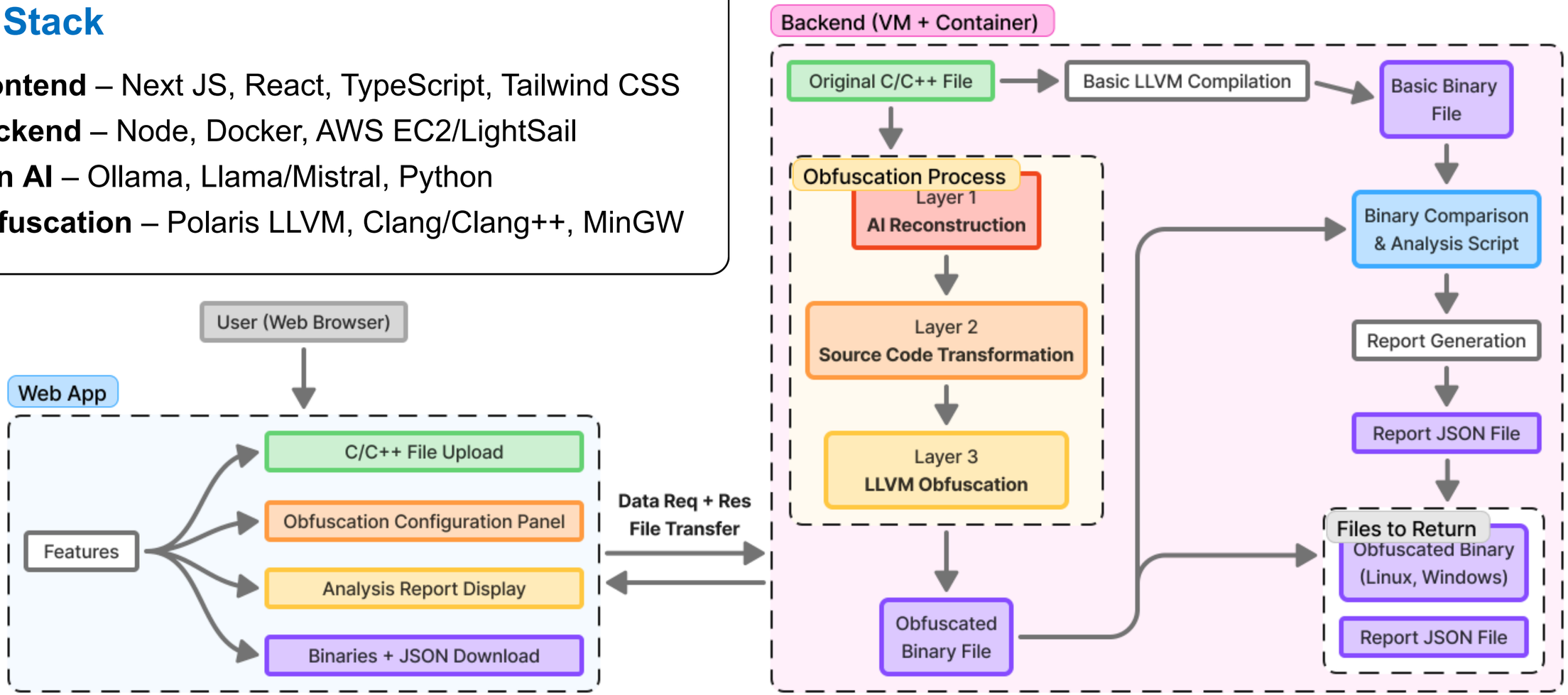
Tech Stack

- **Frontend** – Next JS, React, TypeScript, Tailwind CSS
- **Backend** – Node, Docker, AWS EC2/LightSail
- **Gen AI** – Ollama, Llama/Mistral, Python
- **Obfuscation** – Polaris LLVM, Clang/Clang++, MinGW



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FEASIBILITY AND VIABILITY

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Market Feasibility

- High demanding sectors (software companies, banking, government) & developers **need obfuscation** to **safely distribute executables/binaries**.
- **GUI based interface for budding developers and non-technical professionals.**

Technical Feasibility

- Built on common & high availability Tech stack.
- Gen AI Model can be setup in VM to keep shared code private.
- Uses Open-Source libraries & tools.

Uniqueness & Innovation

- **Only Hybrid solution** combining **AI reconstruction + Source Code Transformation + LLVM Binary Obfuscation.**
- **Quantifies Protection** using **Original & Obfuscated Binaries Comparison**.

Challenges & Solutions

- Large file compilation issues – Timeout with fallback.
- Gen AI availability & cost – Optional Toggle for AI features.
- Gen AI hallucinations – Define Test Cases for code equivalency check.

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IMPACT AND BENEFITS

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Impact

- **Protects proprietary algorithms** from competitors and patent theft.
- Safeguards critical infrastructure code / military software.
- Developers can **monetize software without fear of source code leaking / bypass**.

Benefits

- **Excessive Time Increase to Reverse Engineer** the binary.
- **Multi-Layer Defense** coming from Three Layer Hybrid Obfuscation.
- **NO Installation, CLI or Domain Specific Knowledge Required** making it easy to use for everyone including non-technical professionals.
- Free / Low-Cost to use compared to paid obfuscators.

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RESEARCH AND REFERENCES

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References & Links

- The LLVM Compiler Infrastructure
- Polaris-LLVM Obfuscator
- Obfuscator-LLVM — Software Protection for the Masses
- 11 Code Obfuscation Techniques for Application Protection
- Using LLVM to Obfuscate Your Code During Compilation
- Obfuscator-LLVM